

KSG Revision-4 M1a Composite-v6 Portability Boundary

Correcting a wrong-lane report–figure comparison

Sepehr Mahmoudian

18 August 2026

Abstract

Published C5 commit `be862b155d710573ec95356fc1cbe9a96a2b83b9` failed its first hosted qualification. CodeQL passed. Repository CI had a terminal failed publication job while its terminal 45-job roster completed with 44 successes and one failure; the dedicated composite-v5 route also recorded publication-gate failure. The dedicated log reported that an embedded Form differed from a standalone figure. The v5 checker had compared every rebuilt and committed report with every rebuilt and committed figure, including cross-lane pairs that the TeX source never referenced. This report leaves R5 permanently unissued, defines a direct-child C6 correction using keyed association, and permits R6 only after fresh exact-C6 local closure plus hosted attempt-1 all-success qualification. It changes no v5 publication byte, theorem, estimator, PID object, or numerical result.

Disposition. C5 remains the immutable published subject. Its failed attempt is retained without reinterpretation. R5 is unissued. C6 corrects one checker relation; it does not certify the underlying publication. R6 requires a new exact commit and fresh evidence. This record grants no science, mathematics, security, application, authentication, independence, PDF/UA, or renderer-independence credit.

1 Attempt-1 decision

For exact C5, define

$$Q_5 = L_5 \wedge CI_5 \wedge \text{CodeQL}_5 \wedge D_5, \quad (1)$$

where L_5 is one fresh local qualification observation for exact C5, with no attempt-number or first-attempt authority. The hosted CI_5 , CodeQL_5 , and D_5 terms require terminal success at attempt 1 for that same commit; D_5 is the dedicated composite-v5 route. The hosted observation is

$$D_5 = \text{false}. \quad (2)$$

Hence $Q_5 = \text{false}$, regardless of any other term. The frozen rule

$$\text{issue}(R_5) \iff Q_5 \quad (3)$$

leaves R5 permanently unissued. A retry is attempt 2; changing C5 creates another commit. Neither can retroactively satisfy Equation 1.

2 Terminal hosted observations

Route	Run / result	Bounded observation
Repository CI	32107469096; failed	Terminal roster: 45 jobs, 44 success, 1 failure. Sole failed job 95619717365, step 11, recorded the immutable-v4 mismatch.
CodeQL	32107469060; passed	Four jobs passed; this establishes only the CodeQL term.
Dedicated v5	32107469077; failed	Job 95619716898 passed the C4 failure-surface rechecks, then failed in publication step 8.

Table 1: Exact C5 hosted observations. Provider labels and identifiers grant no authority or authentication credit.

The exact dedicated diagnostic was *composite-v5 boundary PDF check: PDF structure embedded Form content differs from standalone figure*. Later dedicated steps were skipped. This is enough to make D_5 false. It does not by itself identify a damaged report or prove that every PDF consumer would render the committed bytes correctly.

CI job 95619717365, step 11, emitted the analogous v4 diagnostic: *composite-v4 process PDF check: PDF object structure changed: embedded custody Form content differs from the standalone figure*. The v4 and v5 logs are two observations at the same named Cartesian association-rule failure surface, not a second repair and not evidence that either immutable publication is defective. Dedicated-v5 failure remains the decisive false term for R5. Reproducing both diagnostics through the bounded wrong-lane rule proves neither a unique cause nor the only possible remedy.

3 The Cartesian association failure surface

Let

$$\mathcal{R} = (R_{\text{fresh}}, R_{\text{committed}}), \quad \mathcal{F} = (F_{\text{fresh}}, F_{\text{committed}}). \quad (4)$$

The v5 gate evaluated every member of $\mathcal{R} \times \mathcal{F}$ and demanded embedded-Form content equality for every pair. Both TeX report lanes reference the committed figure. A fresh standalone figure rebuilt by another toolchain may therefore be byte-different without being the figure embedded in either report. The cross-product adds wrong-lane comparisons that are not build associations.

The C6 rule is keyed:

$$A_6 = \{(R_{\text{fresh}}, F_{\text{committed}}), (R_{\text{committed}}, F_{\text{committed}})\}. \quad (5)$$

The gate validates embedded content only on A_6 . It compares F_{fresh} with $F_{\text{committed}}$ separately under bounded text, geometry, font, object-safety, and same-renderer raster predicates. A causal hostile supplies an unreferenced lane and requires rejection if Cartesian substitution is restored.

This correction does not declare byte-different PDFs generally equivalent. It changes only the association used by this versioned publication gate. A separate v6 portability adjudicator applies the keyed rule to both immutable v4 and v5 lanes without editing either one.

4 Direct C5 to C6 topology

C6 is an exact unsigned single-parent direct child of C5. There is no R5 commit between them. C6 retains the failed C5 observation and has exactly the contract's 43-path delta: 21 modifications and 22 additions across bounded operational, replay/pointer, publication, schema/tool, and policy surfaces. It changes no theorem or scientific source. The separate immutable-v4/v5 portability adjudicator is another correlated local check, not an independent authority.

Define

$$Q_6 = L_6 \wedge CI_6 \wedge \text{CodeQL}_6 \wedge D_6, \quad (6)$$

where L_6 is local static, hostile, replay, source-state, and publication closure for exact C6; the other terms are terminal hosted success at attempt 1 for that same commit. A typed local record establishes L_6 only after receipt validation binds exact clean C6 pre/post snapshots, the fixed `just ksg-composite-v6` command, successful exit, bounded stdout/stderr, minimal environment, tool identities, and ordering. Missing or invalid local bytes make L_6 false. The issuance policy is

$$\text{issue}(R6) \iff Q_6. \quad (7)$$

Any false, absent, nonterminal, or wrong-commit term, or any wrong-attempt hosted term, leaves R6 unissued. The local record is not an attempt-number or first-attempt authority.

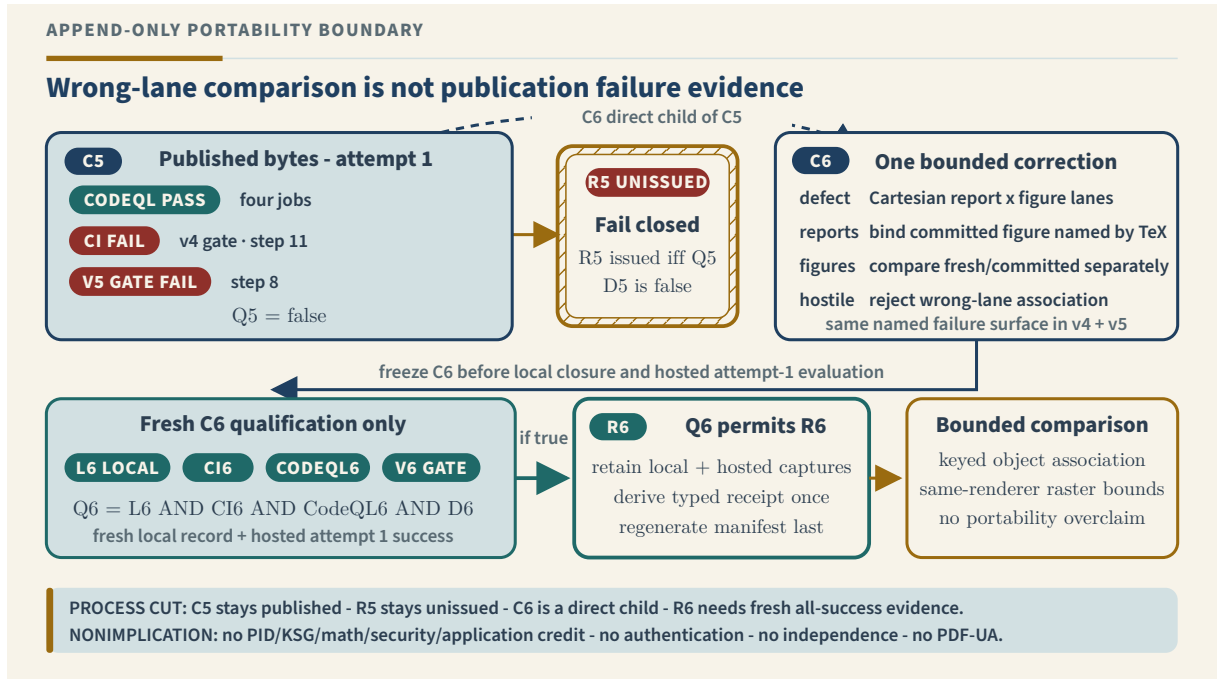


Figure 1: Append-only portability boundary. Shape, line style, text, and color separately distinguish the failed C5 attempt, blocked R5, C6 correction, and conditional R6 route.

When and only when Q_6 is true, R6 has exactly four path changes: the self-excluding source manifest, typed local-closure record, derived receipt, and successor hosted capture. The receipt derives all four terms only after validating the durable local record and fresh exact-C6 hosted capture; the source manifest is regenerated last. Derivation accepts no successor-capture/evidentiary stdin route; it uses distinct stable mode-0600 file descriptors 3 and 4 for local and hosted inputs. These future bytes are not created here. The local record is unsigned and correlated, not authenticated, independent, atomic, or a complete executable inventory. It does not pass ambient variables to the command, and bounded scans reject named secret-like patterns and private-path prefixes; they do not prove privacy.

5 Publication gate and causal controls

The C6 publication set contains an ARIA-labelled SVG with title/description and a TeX text alternative, deterministic standalone figure PDF, four-page report PDF, 120-dpi rendering receipt, 144-dpi visual-review receipt, and their sources. The gate checks exact byte bindings and two isolated same-toolchain rebuilds before applying cross-toolchain structure and raster bounds.

Surface	Fail-closed predicate and causal control
Association	Reports bind only the committed figure referenced by TeX; wrong-lane Cartesian substitution rejects.
Object safety	Unsupported catalog, outline, or page actions; annotations; embedded files; missing resources; nonidentity matrices; inline images; and raster images reachable through Forms, patterns, Type 3 fonts, or soft masks reject.
Placement	Clipping, wrong page, off-page placement, zero scale, rotation, altered page boxes, and altered user units reject.
Text and fonts	Closed headings/body order, embedded subset fonts, encodings, widths, Unicode maps, programs, and report/figure font inventories are checked.
Raster	Color and grayscale comparisons use one local Poppler with explicit bounds; a visible overlay rejects.
Receipt	Exact render rows, hashes, sizes, review pages, and ordered prose are closed; extra claims reject.

Table 2: The hostiles reach the named predicate after outer byte bindings are resealed where needed. A checksum-only rejection receives no causal-test credit.

Same-renderer comparison is correlated differential evidence. It is not renderer independence, PDF/UA conformance, absolute visual correctness, or a general portability theorem.

The clean endpoints use ordinary Git status plus selected metadata checks; rejecting `core.excludesFile` removes one ignore-routing overlay, but repository-ignored products and uninspected Git metadata remain outside the observation and may remain side inputs, so this is not a hermetic closure.

6 Scope boundary

Composite-v6 adds zero PID theories, zero PID functionals, zero estimators, zero theorem-source changes, and zero numerical-result changes. Nothing transfers among categorical or continuous shared exclusions, KSG mutual information, Williams–Beer $I_{\min}$, PID2, PID3, quantized, or mixed-support routes.

Hash equality binds named bytes only. This report establishes no authentication, attestation, authorship, provider completeness, trusted time, peer review, dependency-disjoint independence, scientific or mathematical validity, security certification, privacy assurance, application fitness, safety, control authority, archival preservation, or evidence for any drone or other ecosystem deployment.